

1 Acids Strength

EXPERIMENT 1

Comparison of methods for measuring PH

Goal

The goal of this experiment is to compare three different techniques to measure PH: litmus paper, PH indicators, and PH meters. The overall goal is also to use these techniques as footprints to identify an unknown acid or base.

Materials

- | | |
|---|---|
| ☐ 5 jumbo testube and rack | ☐ 0.1 M H_3PO_4 , 0.1 M CH_3COOH , 0.1 M NaH_2PO_4 , 0.1 M $\text{Zn}(\text{NO}_3)_2$, and 0.1 M NH_4NO_3 |
| ☐ 100mL, 50mL beaker | ☐ Distilled water and tap water |
| ☐ Indicator solutions | ☐ Unknown solution |

Background

Acids are considered dangerous substances and while it is true that special care must be taken when handling some of these products, many acids are used in daily life with great simplicity. For example, aspirin's active principle ingredient is acetylsalicylic acid, vinegar is a 5% solution of acetic acid, and vitamin C is ascorbic acid.

Acids are defined by Brønsted-Lowry as proton (H^+) donors because when dissolved in water, acids dissociate by releasing protons. Strong acids dissociate completely into two ions: the anion is the acid conjugate base and the the cation is a proton. For stoichiometric reasons, the concentration of each ion in the solution equals the initial concentration of the acid. Weak acids do not dissociate completely but, instead, only do so to a certain extent. The strength of weak acids depends on how much they dissociate, a ratio characterized by the acid dissociation constant, K_a .

For a certain weak acid



The equilibrium can be characterized by the expression:

$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

where K_a (capital letter) is the dissociation constant and the brackets represent the concentrations of the respective species.

pH

The acidity depends on the concentration of H^+ in the solution, which will always be a value between 10^{-14} and 1. To avoid using such a large and complex range, a logarithmic scale is preferred, ranging from 0 to 14.

This scale is called the pH, defined by Danish biochemist Sørensen in 1909 as the "power of Hydrogen".

$$\text{pH} = -\log[\text{H}^+]$$

Considering that the strength of acids is their power to generate H^+ , and to compare the strength of different acids, pH should be measured.

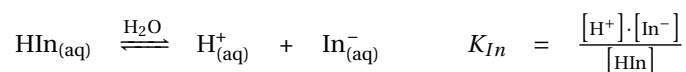
In this experiment, the pH for different solutions will be measured using three different approaches; indicators, pH paper, and a pH meter. Each of these techniques has a different accuracy. While using **pH indicators** it is only possible to determine a range for the pH, **pH paper** will provide an estimated unit, and the **pH-meter** will read down to the decimals.

Indicators

There are several techniques to locate the endpoint during titration. On one hand, we can use a PH meter and measure the PH during a titration. The endpoint locates between the initial and final plateau. We can also use an automatic titrator, a machine that automatically mixes the acid and the base while displaying the PH. Another technique uses an indicator that changes color when the PH changes from acidic to basic or the opposite.

How do indicators function

Acid-base indicators are water-soluble organic dyes that have a different color depending on PH. Halochromism is the effect of changing color due to changes in an external force such as PH. Many flowers or plants exhibit this quality. For example, the french hydrangea is a plant that changes color depending on the acidity of the soil. Turmeric is a specie that turns from yellow to brown when the PH increases. indicator can be extracted from many plants and fruits such as red cabbage, geranium, poppy, rose petals, or blueberries. Indeed, indicators are weak acids that exist in a conjugate base form as well. Both the acid (the indicator form with the hydrogen attached) and the conjugate base have different colors due to differences in their electronic structure that transform into differences in absorbing light. For example, phenolphthalein is a common acid-base indicator that is transparent when the PH is acidic and pink when the PH is basic. The equivalency point of a titration (when acid and base are neutralized) is not necessarily the endpoint (when the indicator changes color). However, if both points are close enough chemical indicators are an accurate way to visually locate the equivalency point during a titration. As indicators are weak acids, each particular indicator is characterized by a constant of acidity referred to as K_{In} so that



The color change is considered to happen when the difference between the acidic and basic form of the indicator is an order or magnitude—ten times—different. This is the ten times rule. For this difference, we have that

$$PH = PK_{In} \pm 1$$

We can always use a different value for the difference (Δ) between the acid and basic form of an indicator so that the PH range will be given by

$$PH = PK_{In} \pm \log(\Delta)$$

This expression helps with the selection of the indicator for a given titration. For example, if we want to select an indicator that changes color when the difference between acid and base form is 5 times ($\Delta = 5$), then the OH range would be: $PH = PK_{In} \pm 0.7$

Selecting an indicator

A good indicator will turn into a different color within the transition between acid and basic PH during titration. For example, for the titration of hydrochloric acid with sodium hydroxide, the PH changes from 5 before the equivalency point to 8 after. Any indicator with PK_{In} between 5 and 8 would be able to accurately indicate the stoichiometric point. Examples are bromothymol blue, m-nitrophenol, and even phenolphthalein which transition color at 8. For the titration of a weak acid and a strong base, the selection of an indicator is more difficult as the difference between the acidic and basic plateaus during the titration is less pronounced. We must choose an indicator that changes color more closely to the endpoint. Table 1 reports numerous acid-base indicators and their colors.

Litmus paper

Litmus paper is an indicator heavily used for quick PH testing in labs. It contains a mixture of dyes that were extracted originally from lichens. There are two main types of litmus paper, red and blue litmus. Light blue litmus paper turns red when the medium is acidic, whereas red litmus paper turns blue in basic mediums. By wetting a piece of litmus paper one can also test for the acidity of gases, for example, ammonia vapor would turn blue on a piece of wet red litmus paper.

Table 1 Acid-base indicators showing ample representations of their colors

Name	PH Range	Color acid form		Color basic form
Thymol Blue	1.2-2.8	red		yellow
Pentamethoxy red	1.2-2.3	red-violet		colorless
Methyl yellow	2.9-4.0	red		yellow
Methyl orange	3.1-4.4	red		orange
Bromophenol blue	3.0-4.6	yellow		blue-violet
Tetrabromophenol blue	3.0-4.6	yellow		blue
Alizarin sodium sulfonate	3.7-5.2	yellow		violet
α -Naphthyl red	3.7-5.0	red		yellow
p-Ethoxychrysoidine	3.5-5.5	red		yellow
Bromocresol green	4.0-5.6	yellow		blue
Methyl red	4.4-6.2	red		yellow
Bromocresol purple	5.2-6.8	yellow		purple
Bromophenol blue	6.2-7.6	yellow		blue
Azolitmin	5.0-8.0	red		blue
Phenol red	6.4-8.0	yellow		red
Neutral red	6.8-8.0	red		yellow
Rosolic acid	6.8-8.0	yellow		red
Cresol red	7.2-8.8	yellow		red
α -Naphtholphthalein	7.3-8.7	rose		green
Tropeolin	7.6-8.9	yellow		rose-red
Thymol blue	8.0-9.6	yellow		blue
Phenolphthalein	8.0-10.0	colorless		red
α -Naphtholbenzein	9.0-11.0	yellow		blue
Thymolphthalein	9.4-10.6	colorless		blue
Nile blue	10.1-11.1	blue		red
Diazo violet	10.1-12.0	yellow		violet
Tropeolin	11.0-13.0	yellow		orange-brown
Nitramine	11.0-13.0	colorless		orange-brown
Trinitrobenzoic acid	12.0-13.4	colorless		orange-red

The experiment

This experiment is divided into the following parts:

- Determining pH range using indicators.
- Determining pH using pH paper.
- Determining pH using a pH meter.

Part A. Determining pH range using indicators.

A pH indicator is a chemical substance that, when added to a solution, changes its color depending on the solution's pH. The pH at which color changes is called the **equivalence point**. Some indicators have one equivalence point while others have more than one equivalence points. By using an indicator it will be possible to determine a pH range only. Combining different indicators the range can be narrowed down.

Example

We test a sample with three different indicators by dividing the sample into three small samples. Each small sample is tested using an indicator from the following list: i) Indicator A is colorless for pH values lower than 10 and turns pink for a pH greater than 10. ii) Indicator B is blue up to pH 3, then it turns colorless. iii) Indicator C is colorless for a pH lower than 6, when it turns red. The red color changes to yellow between pH 9 and pH 10, and stays yellow for higher pH values. Estimate the pH range for the solution if the three samples are colorless.

Answer: Indicator A tells that the solution's pH is lower than 10. Therefore, the range must be $0 < \text{pH} < 10$. Indicator B shows that the solution's pH is greater than 3 (3 to 14). This information, combined with that of indicator A, narrows the pH range to $3 < \text{pH} < 10$. Indicator C tells that the solution's pH is lower than 5. The final range is $3 < \text{pH} < 5$.

Parts B. Determining pH using pH-paper.

pH paper, or **universal pH paper** is a strip of paper impregnated with a mixture of indicators. The paper changes color smoothly for each pH value and the pH is determined by matching the color of the paper with that in a chart attached to the pH paper. In the best case, by using pH paper it will be possible to select one value for the pH, although in some situations when the color is not one in the chart, it will be possible to indicate a range of two numbers only.

Parts C. Determining pH using a pH-meter.

pH meters are scientific instruments that measure the pH of solutions. The tool has a probe that is inserted in the liquid and measures the concentration of protons based on the voltage between two electrodes. Their use is similar to that of a conductivity-meter, and some instruments can perform both tasks. Probes need to be cleaned after each measurement to avoid contamination and should never be touched with the hands or paper towels. Calibration of the tool with solutions of known pH ensures the best accuracy. After use, probes are best kept moist, typically in distilled water. A pH meter can offer a precision down to the hundredth of the unit of pH.

Procedure

Part A. Determining pH range using indicators.

- ☐ *Step 1:* – Obtain 5 jumbo test tubes and a rack. Clean the test tubes and let them drain upside down in the rack.
- ☐ *Step 2:* – Obtain a 100 mL beaker with plenty of distilled water and transfer 4.0 mL in a graduated cylinder.
- ☐ *Step 3:* – Label one test tube with the letter R, and pour the 4.0 mL of distilled water into it. This test tube will be used as a volume reference for the following test tubes.
- ☐ *Step 4:* – In a 50 mL beaker get 40 mL of one of the following list of solutions (you will use the others further in the experiment). These solutions are going to be referred to here as the acid or base solutions.

- | | | | |
|-----------------------------------|-------------------------------------|-----------------------------------|---------------------|
| 1. 0.1 M H_3PO_4 | 3. 0.1 M NaH_2PO_4 | 5. 0.1 M NH_4NO_3 | 7. Tap water |
| 2. 0.1 M CH_3COOH | 4. 0.1 M $\text{Zn}(\text{NO}_3)_2$ | 6. Distilled water | 8. Unknown solution |

CAUTION!

-  Handle concentrated acids with care to avoid chemical burns.
-  Are you wearing your goggles? Do you know where the eye-washer is?

- ☐ *Step 5:* – Obtain a 250 mL beaker for waste, a medicine dropper or a plastic pipette
- ☐ *Step 6:* – Label the rest of the test tubes from 1 to 4. Use test tube R as a reference for the volume to add in the following test tubes. Rinse the dropper with the distilled water (discarding it in the waste beaker) between different solutions. Add 4.0 mL of the solution chosen to each test tube.
- ☐ *Step 7:* – Add two drops of Thymol Blue to test tube 1. Using the color table provided in the background section, indicate the PH range of the solution.
- ☐ *Step 8:* – Add two drops of Methyl Orange to test tube 2. Using the color table provided in the background section, indicate the PH range of the solution.
- ☐ *Step 9:* – Add two drops of Methyl Red to test tube 3. Using the color table provided in the background section, indicate the PH range of the solution.

- ☐ *Step 10:* – Add two drops of Bromothymol Blue to test tube 4. Using the color table provided in the background section, indicate the PH range of the solution.
- ☐ *Step 11:* – Record the colors on the results page.
- ☐ *Step 12:* – Reserve the acid or base solution for Parts B and C (the beaker you used to fill the test tubes). Set the beaker on a paper towel and label it.
- ☐ *Step 13:* – Discard the colored solutions tested with the indicator and clean the test tubes.
- ☐ *Step 14:* – Repeat steps 4 to 12 all other solutions given in Step 4. Make sure to rinse the dropper.

Part B. Determining pH using pH-paper.

- ☐ *Step 1:* – Obtain a stirring rod and the universal indicator paper. Be careful not to contaminate the paper with dirty gloves or bare hands. Always set the (small) pieces of paper on a clean paper towel.
- ☐ *Step 2:* – Rinse the stirring rod with the wash bottle over the waste beaker. Rinse thoroughly between different solutions.
- ☐ *Step 3:* – Immerse the tip of the stirring rod in one solution and touch a piece of pH paper with the rod.
- ☐ *Step 4:* – Match the color of the drop on the paper with the color chart on the plastic case. Record the pH value in the results section.
- ☐ *Step 5:* – Repeat the same process; rinsing, dipping, and touching, for the rest of the solutions.
- ☐ *Step 6:* – Reserve the solutions in the beakers for part C.

Part C. Determining pH using a pH-meter.

- ☐ *Step 1:* – Install the pH meter. The probe of the instrument is immersed in a buffer solution that must be kept aside until the end of the experiment.
- ☐ *Step 2:* – Press Standarize (top right menu). Then press Clear and confirm. Save when the measurement is stable at the buffer's PH value (i.e. 10.00). At this point, the PH meter will be standardized.
- ☐ *Step 3:* – Cleaning and measuring. It is very important to follow the cleaning steps before each measurement. You will need a 100 mL beaker labeled waste a wash bottle and a 100 mL beaker with distilled water.
 1. Rinse the probe using the wash bottle over the waste beaker.
 2. Dip the probe 3-4 times into the 250 mL beaker with clean distilled water.
 3. Dip the probe into the solution to be measured. Use a 50 mL beaker with at least 25 mL of solution.
 4. Repeat cleaning steps 1 and 2.
- ☐ *Step 4:* – Following the procedure outlined in the step above, measure the eight solutions.
- ☐ *Step 5:* – Record all the values.
- ☐ *Step 6:* – Make sure you leave the PH meter's probe immersed in the buffer solution, as you originally found it.

STUDENT INFO

Name: _____ Date: _____

Pre-lab Questions

Comparison of methods for measuring PH

1. You add a few drops of tymol blue to a clear solution and its color quickly changes to red. What is the PH range of the solution?
2. You add a few drops of tymol blue to a clear solution and its color quickly changes to orange. What is the PH range of the solution?
3. Name the three methods to determine pH to be used in this laboratory.
4. Order the three methods used in this experiment to calculate PH from most to the least accurate, according to your expectations.
5. We test a sample with three different indicators by dividing the sample into three small samples. Each small sample is tested using an indicator from the following list: i) Indicator A is colorless for pH values lower than 10 and turns pink for a pH greater than 10. ii) Indicator B is blue up to pH 3, then it turns colorless. iii) Indicator C is colorless for a pH lower than 6 when it turns red. The red color changes to yellow between pH 9 and pH 10, and stays yellow for higher pH values. Estimate the pH range for the solution if the samples tested with indicators A and B become colorless and the one tested with indicator C turns reddish.

STUDENT INFO

Name:

Date:

Comparison of methods for measuring PH

Results

EXPERIMENT 1

Solution	Indicators						PH Paper	PH Meter
	Thymol Blue	Methyl Orange	Methyl Red	Bromothymol Blue		PH		
	Color	PH Range	Color	PH Range	Color			
H ₃ PO ₄	_____	_____	_____	_____	_____	_____	_____	_____
CH ₃ COOH	_____	_____	_____	_____	_____	_____	_____	_____
NaH ₂ PO ₄	_____	_____	_____	_____	_____	_____	_____	_____
Zn(NO ₃) ₂	_____	_____	_____	_____	_____	_____	_____	_____
NH ₄ NO ₃	_____	_____	_____	_____	_____	_____	_____	_____
Distilled water	_____	_____	_____	_____	_____	_____	_____	_____
Tap water	_____	_____	_____	_____	_____	_____	_____	_____
Unknown	_____	_____	_____	_____	_____	_____	_____	_____
Unknown # _____							Unknown Identity _____	

STUDENT INFO

Name: _____ Date: _____

Post-lab Questions

Comparison of methods for measuring PH

1. Based on your observation, identify the unknown.
2. Based on your observation, which of the three techniques is the most accurate?
3. Calculate the concentration of H_3O^+ ions in the most acidic solution, using the pH-meter value.
4. Calculate the concentration of H_3O^+ ions in the less acidic solution, using the pH-meter value.

